



Evaluating Hybrid Deep Learning Architectures for Day-Ahead Electricity Price Forecasting in the German Spot Market

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Introduction



Motivation & Problem Context



Motivation

- High socioeconomic & geopolitical dependence on electricity production in Germany
- Increasing electricity prices:
 - Positive correlation with consumer prices
 - Negative correlation with GDP growth [1]
- Knowing prices beforehand allows for more accurate purchasing / budgeting

Problem

- Frequent regime changes from socioeconomic instability
 - COVID-19 pandemic [2]
 - Russo-Ukraine War [3]
- Inherent extreme shocks [2]
- Dependent on a multitude of unforeseeable variables
 - e.g., environmental, logistical, regulatory [4]
 - Highly non-linear dependencies [2]

Solution

- Two-step hybrid framework
 - Signal decomposition to simplify signals
 - Deep learning to model each subsignal separately
- Specialized artificial neural network (ANN) architecture to learn both local and long-term dependencies



Methods



Overview of the German Day-Ahead Electricity Spot Market Data



Sources

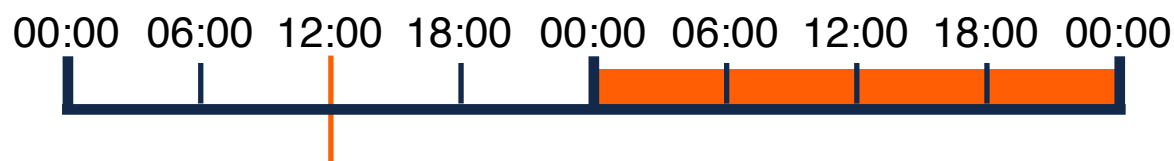
- Weather data from *Bright Sky API* [5]
- Electricity-related data from *Energy-Charts API* [6]

Scope

- Collected hourly data from 2019 through 2024
- Averaged weather data across Berlin, Hamburg, Munich, and Cologne

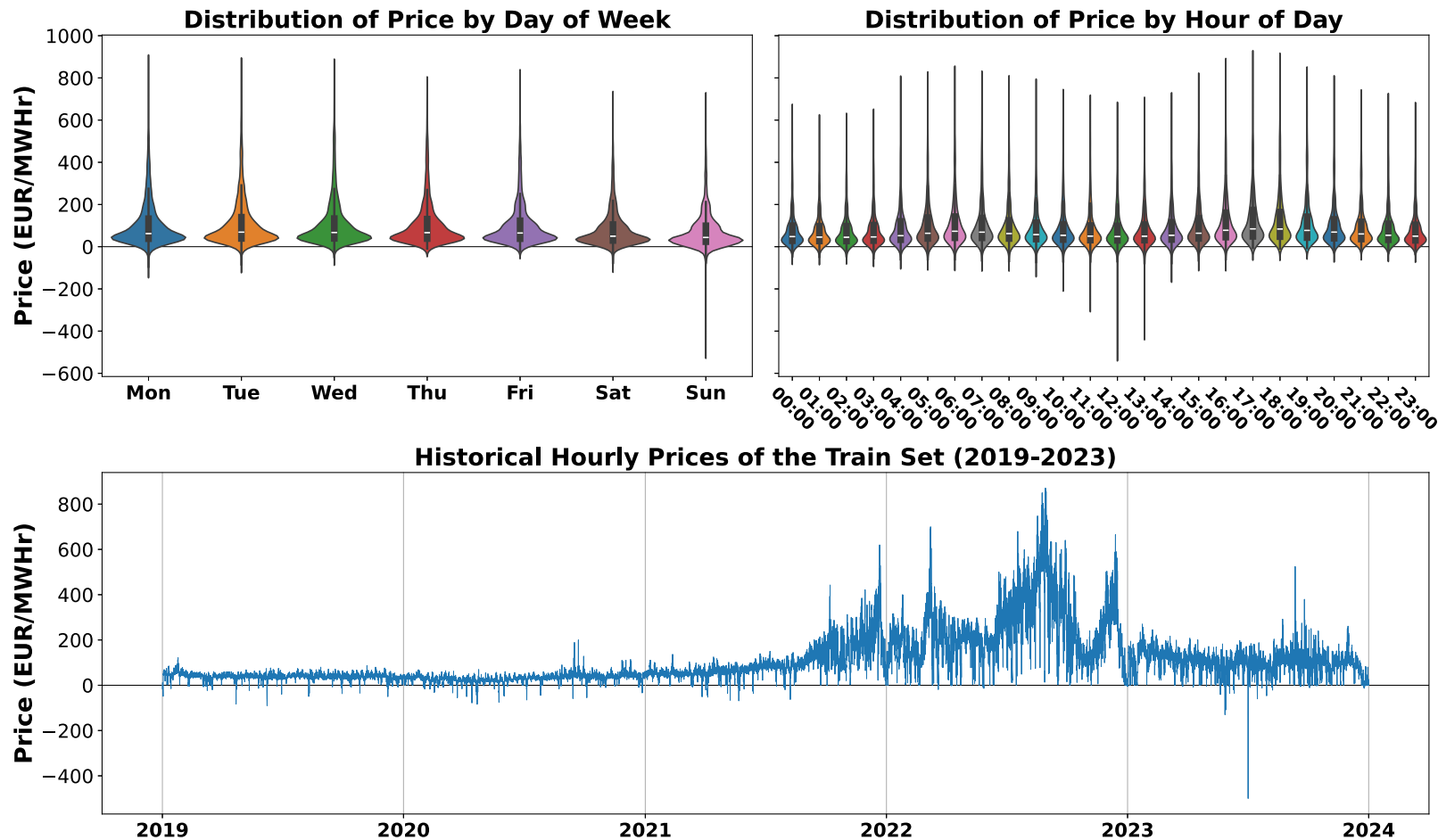
Constraints

- Prices for each hour of next day set at noon the day before [7]



Feature	Units
Precipitation	mm
Temperature	°C
Humidity	%
Generation from Renewables	MW
Load	MW
Load from Renewables	MW
Cross-Border Electricity Trade Volume	GW
Daily / Weekly / Monthly Rolling Price Mean	EUR
Daily / Weekly / Monthly Rolling Price Variance	EUR ²

Distributional Characteristics & Temporal Dynamics of Historical Electricity Prices



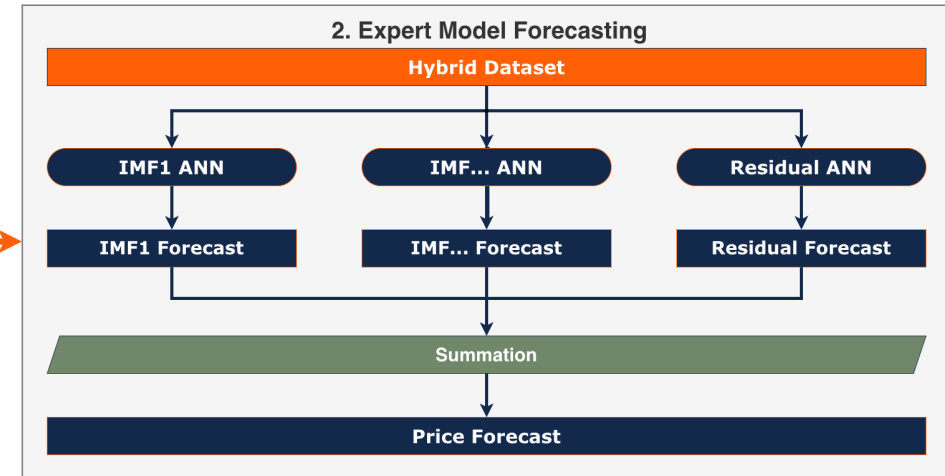
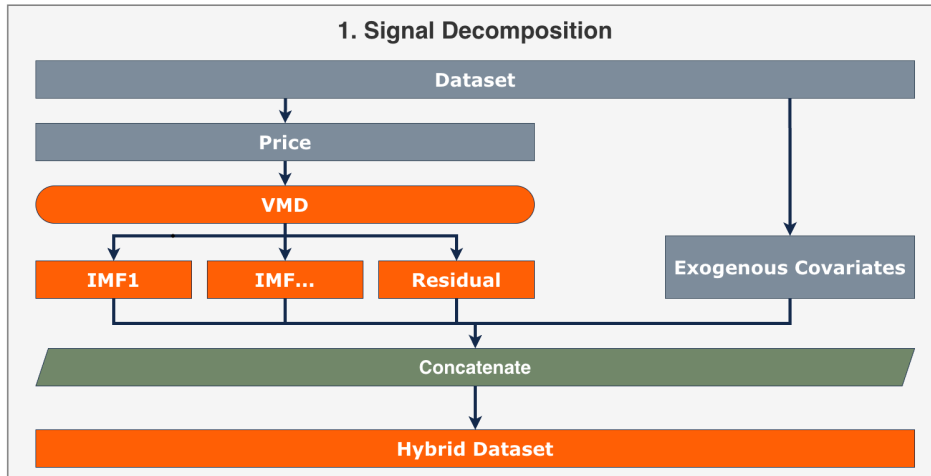
Seasonality

- Inter-day seasonality
- Mon—Wed have peak price
- Sat—Sun have the lowest price
- Intra-day seasonality
- Morning / evening have the highest prices throughout the day

Historical Behavior

- Extreme shocks in price apparent
- Frequent regime changes
- Regime switches during the Russo-Ukraine War
- General upward trend
- Increased variability over time

Proposed Hybrid Forecasting Framework



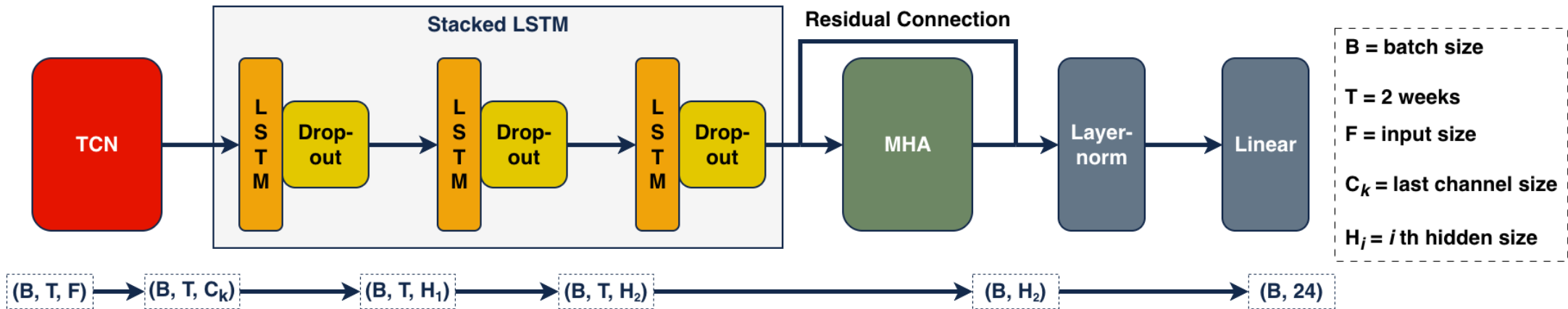
Signal Decomposition

- Decompose using Intrinsic Mode Functions (IMFs) using Variational Mode Decomposition (VMD) [8]
- Each IMF represents a different level of frequency, complexity, and seasonality of price [9]
- Improved learning of complex, multi-seasonality [10]
- Frequency separation analysis to preselect the number of IMFs for VMD to produce [11]

Expert Model Forecasting

- A different ANN is trained on each IMF, learning behaviors specific to that frequency [9]
- Expert ANNs forecast their own IMFs
- IMF forecasts are summed for the price forecast
- The residual (the difference between the raw signal and the summed IMFs) is also modeled

Proposed TCN-LSTM-MHA Network Architecture



Temporal Convolutional Network (TCN)

- Capture local, seasonal, and sub-seasonal dependencies within the data [12]
- Acts as a feature extractor for stacked LSTM
- Allows LSTM stack to learn long-term patterns [13]

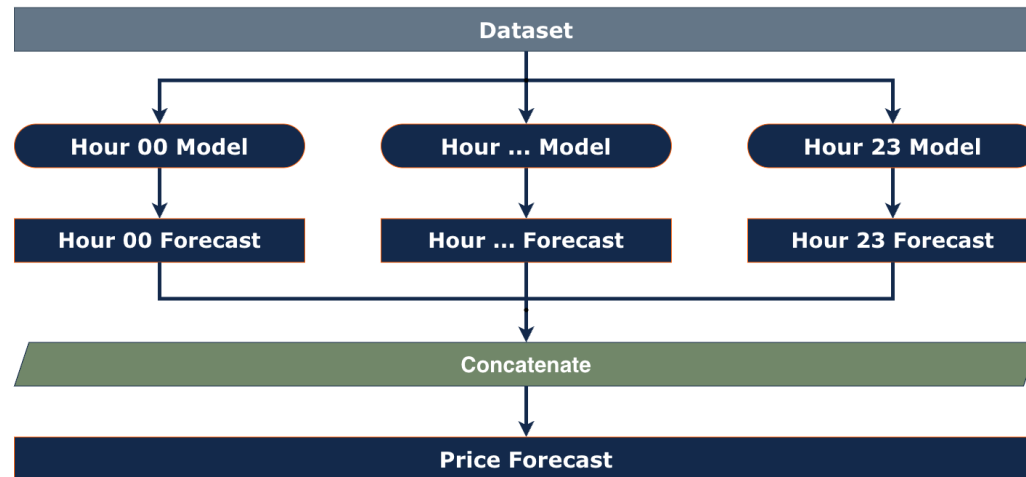
Stacked Long Short-Term Memory (LSTM)

- Learns longer-term relationships and structure [14]
- Gated unit structure aids in handling nonstationarity and regime-shifting structure [15]

Multi-Head Attention (MHA)

- Heads learn lagged correlations [16]
- Provides robustness for irregular events [17]

Benchmark & Alternative Forecasting Frameworks



Direct Multi-Step Forecasting (DMF)

- Individual machine learning (ML) models learn and forecast each hour of the next day, separately
- Effectively allows sequence-to-sequence forecasts
- DMF models are provided with data from noon of the previous day to noon of the current day

Class

Model

DMF

- Elastic Net (EN)
- Light Gradient Boosting Machine (LGBM)
- Linear Regression (LR)
- Random Forest Regressor (RFR)
- Support Vector Regression (SVR)
- EXtreme Gradient Boosting Regressor (XGBR)

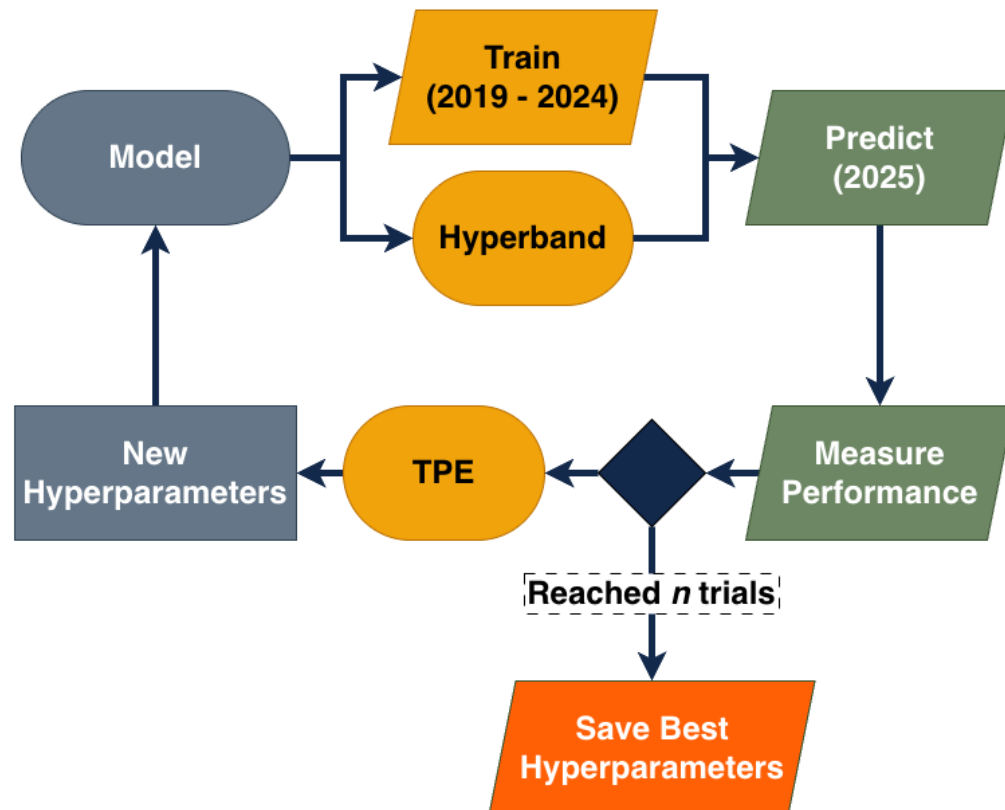
Stand-alone ANN

- TCN-LSTM-MHA
- TCN-MHA
- LSTM-MHA
- TCN-LSTM
- TCN
- LSTM

Hybrid

- VMD-TCN-LSTM-MHA (VTLM)
- VMD-TCN-MHA (VTM)
- VMD-LSTM-MHA (VLM)
- VMD-TCN-LSTM (VTL)
- VMD-TCN (VT)
- VMD-LSTM (VL)

Hyperparameter Optimization & Temporal Validation



Hyperparameter Tuning

- Bayesian Optimization (BO) framework using Tree-Structured Parzen Estimator (TPE)
- TPE is nonparametric and searches high-dimensional, conditional parameter spaces better than alternatives [18]
- Hyperband pruning discards unpromising trials early during training (ANNs only) [19]
- Early stopping terminates training after unimproved validation performance for 5 epochs (ANNS only) [20]

Cross-Validation (CV)

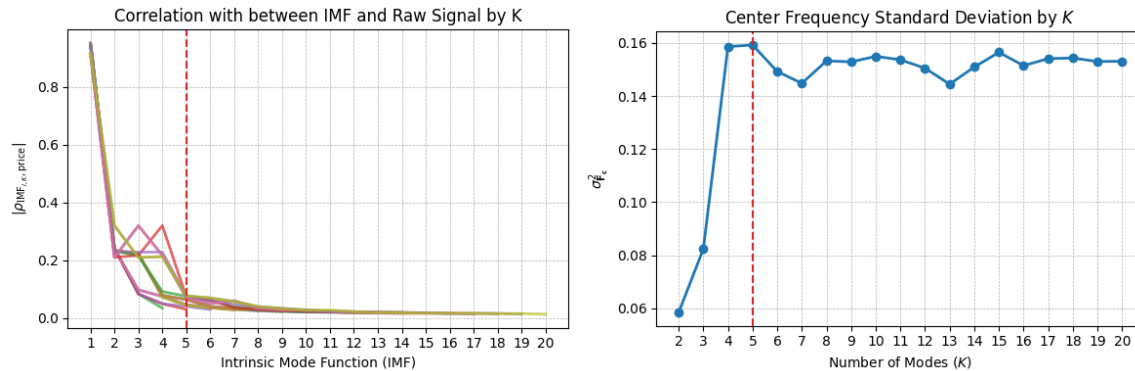
- Rolling window temporal CV with 5 folds was used in the hyperparameter tuning pipeline for DMF models
- No CV used for ANNs due to computational constraints



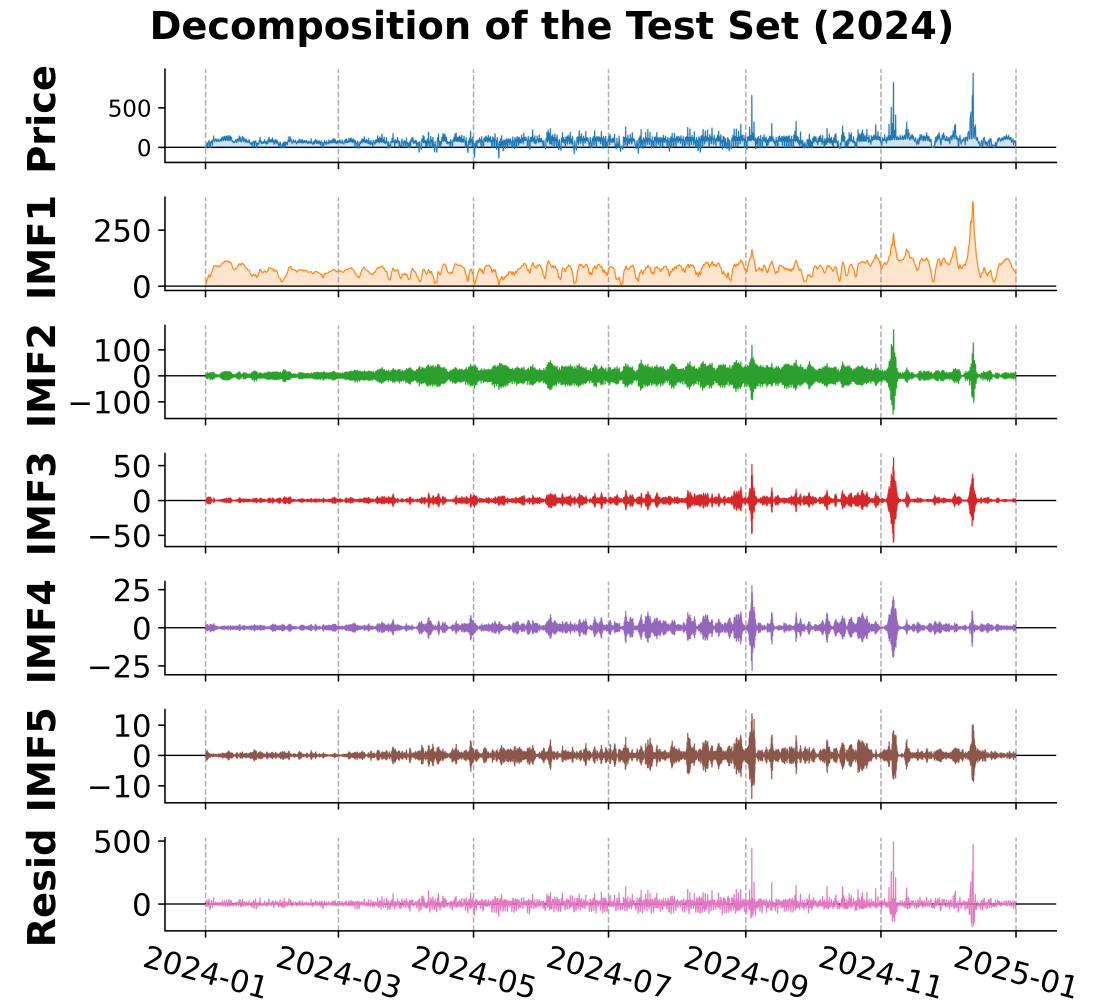
Results



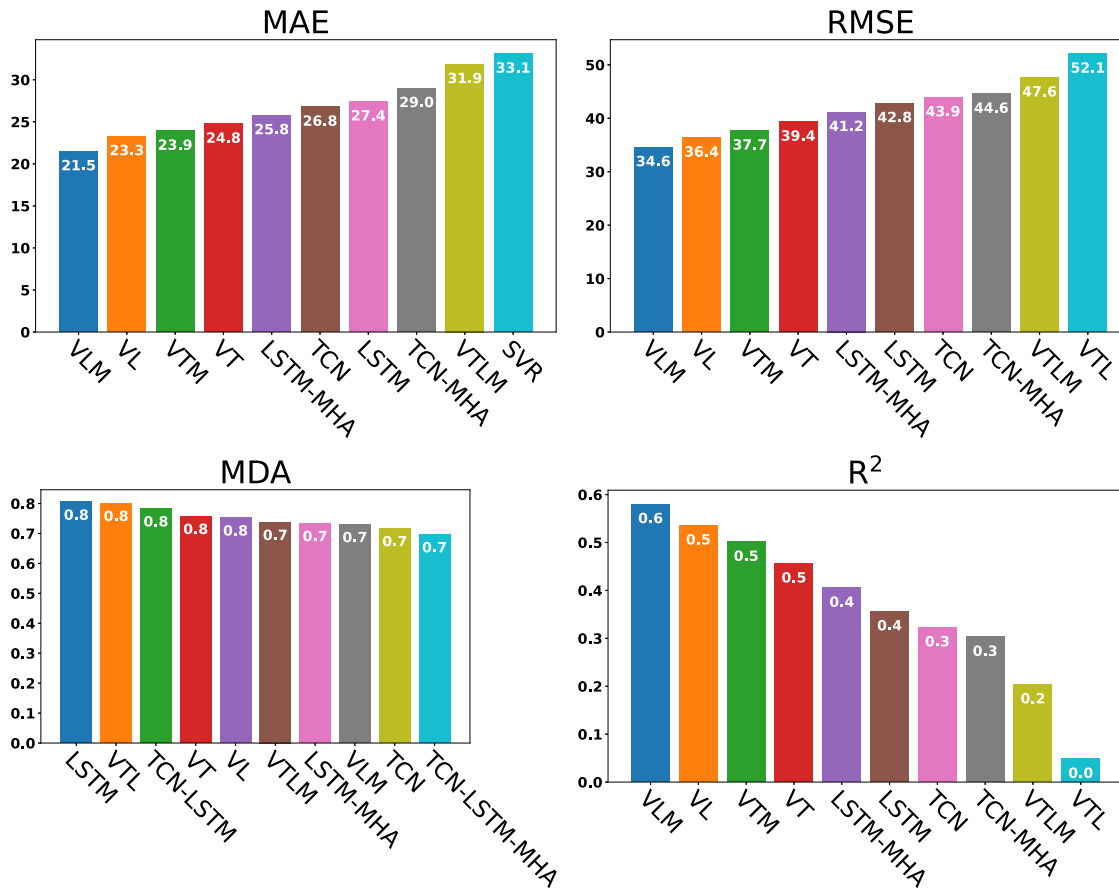
Frequency-Based Analysis for VMD Hyperparameter Selection



- Tested many mode numbers, $k \in \{2, 3, \dots, 20\}$
- The highest standard deviation of center frequencies was found at $k = 5$, $\alpha = 4,000$
- Correlation tended to dip after IMF5 for all values of k
- Overall, IMFs tended to represent different frequencies
- IMF1 captured the general trend of historical prices
- Substantial mode mixing present, although this was common across all tested decompositions



Test Set Forecast Accuracy Metrics



Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Directional Accuracy (MDA)

$$MDA = \frac{1}{n-1} \sum_{i=2}^n \mathbf{1}_{\{(y_i - y_{i-1})(\hat{y}_i - \hat{y}_{i-1}) > 0\}}$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

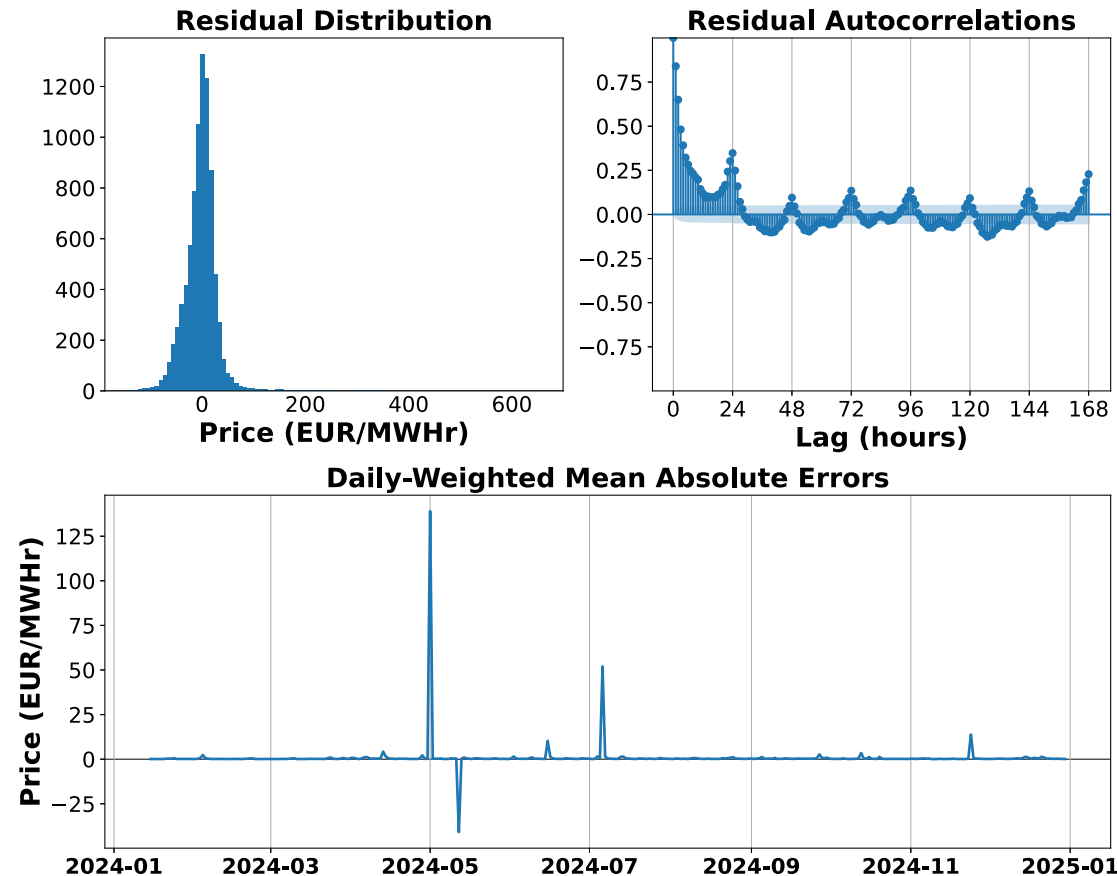
Coefficient of Determination (R²)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

(Only the top ten models are shown for each metric.)

- VMD-LSTM-MHA (VLM) architecture demonstrated the lowest MAE, RMSE and the greatest MDA, R²
 - MAE = 21.502 EUR
 - RMSE = 34.604 EUR
 - MDA = 0.731 = 73.1 %
 - R² = 0.580 = 58 %

Test Set Forecast Accuracy of the VMD-LSTM-MHA Model



Residual Distribution

- Residuals are approximately centered around zero
- Long right skew implies rare but drastic under-prediction

Residual Autocorrelation Function

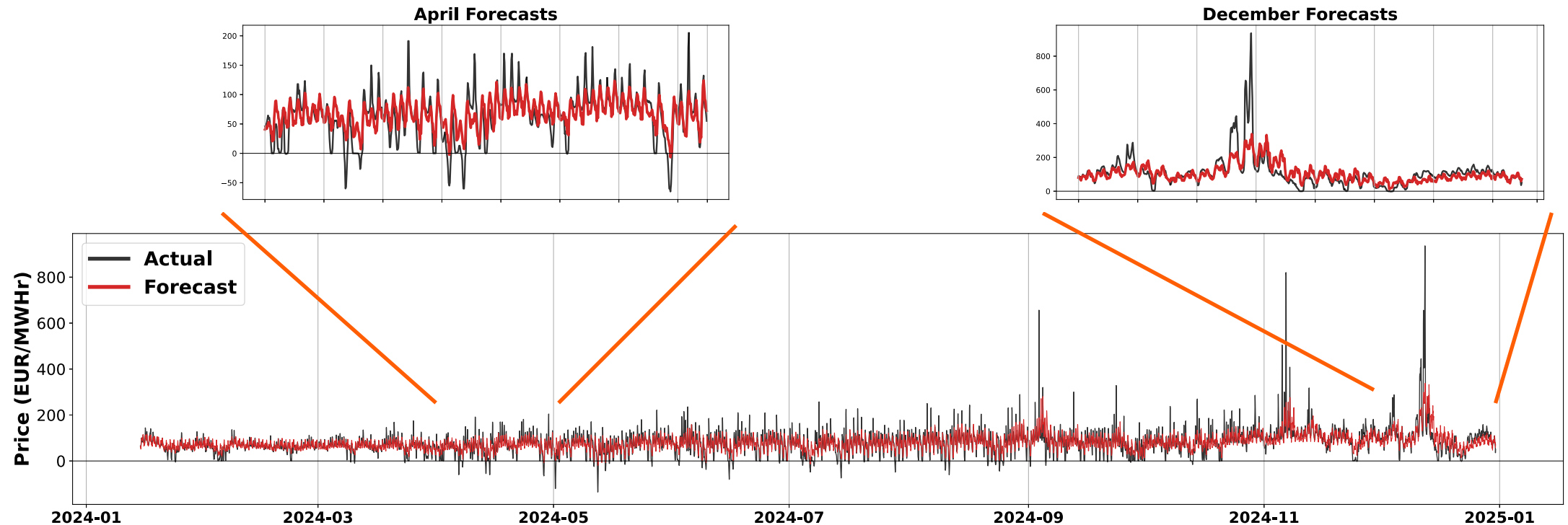
- Significant autocorrelation in residuals
- Implies a large amount of accounted lagged feature importance in network weights

Daily-Weighted Mean Absolute Errors

$$\text{DMAE} = \frac{1}{24} \sum_{i=1}^{24} \frac{|y_i - \hat{y}_i|}{\bar{y}_{24}}, \text{ i.e. } \bar{y}_{24} \text{ is the daily mean}$$

- Demonstrates performance relative to daily market conditions, or difficulty-adjusted error [7]
- ~5 days of extremely poor relative forecast accuracy

Actual vs. Forecasted Prices for the VMD-LSTM-MHA Model (2024)



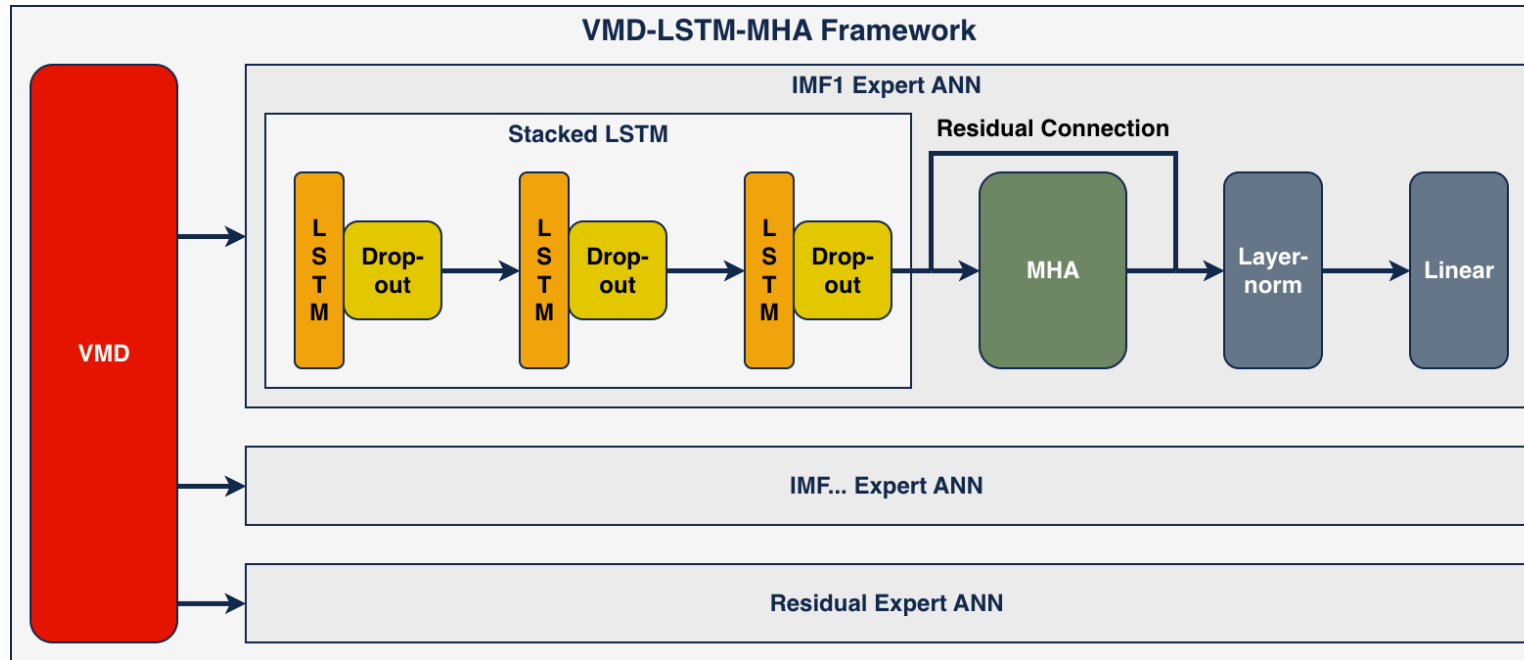
- The VMD-LSTM-MHA network tends to follow intra-day and inter-day seasonality well
- Consistent under-prediction during regular periods
- Under-prediction of extremity of shocks, although some indication of increased price variance is present
- Diebold-Mariano test rejects the null hypothesis; forecasts are more accurate than 2nd best model's [21]



Discussion



Key Findings from the VMD-LSTM-MHA Model



- Best performing model (lowest MAE, RMSE and highest R^2 , MDA) was chosen to be VMD-LSTM-MHA
- Excludes TCN, suggesting a substantial amount of local, sub-seasonal noise in day-ahead electricity price data
- VMD was shown to deconstruct the complex, original price signal into significantly more interpretable IMFs
- MHA significantly improved model performance, likely due to the inherent irregularities of electricity prices

Strengths, Limitations, & Future Work



Strengths

- Highly structured, intentional neural network design
- Tested most combinations of network layers for robustness
- Robust analysis of frequency separation to guide VMD [11]
- VMD-LSTM-MHA bypasses the need for lagged feature selection [16]
- Model frameworks completely abide by real-time, day-ahead trading constraints [7]

Limitations

- Mode mixing between IMFs could signify still imperfect separation
- The test set largely includes only one price regime
- Residual ACF shows lagged correlation structure not accounted for by MHA
- Expert DMF models miss the hourly covariance structure
- Expert DMF models may struggle during fitting due to high dimensionality

Future Work

- Incorporate automated feature selection for DMF models
- Ordinary Least Squares (OLS) regression on expert hybrid model forecasts to correct forecasts
- Exploration of transformer-based architectures (e.g., Temporal Fusion Transformer (TFT))
- Further exploration of probabilistic forecasting methods (e.g., Conformal Predictions (CP))

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Thank You



Questions?

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